

Web Engineering & Development (SWE 363)

4.1 JavaScript Fundamentals

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In today's lecture:

- Objects
- Maps
- String
- Date
- Math
- Exception Handling
- Regular Expressions

Reference:

- Zybook: 4.9 to 4.15

Announcements

- Project Phase #2 is due This Thursday at 11:59 PM
- Quiz next class

object Examples

```
const fruits = { "f1": "apple", "f2": "grape", "f3": "banana", "f4": "peach" };
const keys = Object.keys(fruits);
for (let i = 0; i < keys.length; i++) {
    console.log(keys[i], "=", fruits[keys[i]]);
}
```

```
const fruits = { "f1": "apple", "f2": "grape", "f3": "banana", "f4": "peach" };
let i = 0;
while (i < keys.length) {
    console.log(keys[i], "=", fruits[keys[i]]);
    i++;
}
```

```
const fruits = { "f1": "apple", "f2": "grape", "f3": "banana", "f4": "peach" };
const keys = Object.keys(fruits);
let j = 0;
do {
    console.log(keys[j], "=", fruits[keys[j]]);
    j++;
} while (j < keys.length);
```

```
for (let k in fruits) {
    console.log(k, "=", fruits[k]);
}
```

```
const fruits = { "f1": "apple", "f2": "grape", "f3": "banana", "f4": "peach" };  
for (let k in fruits) {  
    console.log(k, "=", fruits[k]);  
}
```

```
const fruits = { "f1": "apple", "f2": "grape", "f3": "banana", "f4": "peach" };  
for (let [key, value] of Object.entries(fruits)) {  
    console.log(key, "=", value);  
}
```

Modern alternative methods and properties

```
// Create a new Map
let myMap = new Map();

// 1 set(key, value) → add or update a key-value pair
myMap.set("name", "Ali");
myMap.set("age", 21);
console.log(myMap); // Map { 'name' => 'Ali', 'age' => 21 }

// 2 get(key) → retrieve the value by key
console.log(myMap.get("name")); // Ali

// 3 has(key) → check if a key exists
console.log(myMap.has("age")); // true
console.log(myMap.has("grade")); // false

// 4 delete(key) → remove a key-value pair
myMap.delete("age");
console.log(myMap); // Map { 'name' => 'Ali' }

// 5 size → number of elements in the map
console.log(myMap.size); // 1
```

Exception Handling

```
function divide(a, b) {  
  try {  
    if (b === 0) {  
      // throw an error if dividing by zero  
      throw new Error("You cannot divide by zero!");  
    }  
    let result = a / b;  
    console.log("Result:", result);  
  } catch (error) {  
    // handle the error  
    console.log("Error:", error.message);  
  } finally {  
    // this block always runs  
    console.log("Operation finished.");  
  }  
}  
  
// Test cases  
divide(10, 2); // ✅ prints Result: 5  
divide(10, 0); // ❌ prints Error and still runs finally
```


Regular Expressions – Quick Reference with `.test()`

<code>/abc/</code>		<code>/abc/.test("abc123");</code> <code>/abc/.test("xyz");</code>	<code>true</code> <code>false</code>
<code>/^abc/</code>	Matches "abc" at the start	<code>/^abc/.test("abcdef");</code> <code>/^abc/.test("zabc");</code>	<code>true</code> <code>false</code>
<code>/abc\$/</code>	at the end	<code>/abc\$/.test("123abc");</code> <code>/abc\$/.test("abc123");</code>	<code>true</code> <code>false</code>
<code>/a.c/</code>	<code>.</code> matches any one char	<code>/a.c/.test("abc");</code> <code>/a.c/.test("ac");</code>	<code>true</code> <code>false</code>
<code>/a*/</code>	<code>*</code> 0 or more a's	<code>/a*/.test("aaab");</code> <code>/a*/.test("bbb");</code>	<code>true</code> <code>false</code>

45m

Demo

Starter code at: [web-engineering-kfupm-classroom-4-1-js-advanced/](https://github.com/kfupm-classroom/web-engineering-kfupm-classroom-4-1-js-advanced/)

Next Class

- Quiz